Investigation: Exposing Unity Catalog Tables via Iceberg REST Catalog API

> ***ADO Ticket***: *#186438* ***Author***: *Carlos Carrillo (* `ccarrillo@oncologyanalytics.com` *)* ***Date***: *2026-03-23 (updated 2026-03-25)* ***Status***: *COMPLETE — Investigation finished. Iceberg REST endpoint validated; 3 implementation prerequisites identified.* ***Project context***: `../../knowledge.json` *v1.14.0 — operational facts cited as [K].*

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1. Executive Summary

Verdict: GO — The Iceberg REST Catalog API is technically viable for exposing UC tables to external services. Azure Databricks natively supports the endpoint at `/api/2.1/unity-catalog/iceberg-rest`. The API has zero incremental licensing cost. Network connectivity validated. Full workspace inventory captured.

Validation: PAT validated 2026-03-24 against 12 API endpoints including Iceberg REST Catalog. Full UC inventory captured: 8 catalogs, ~66 schemas, ~466 tables. DEV workspace (`adb-3806388400498653`) contains PHI — TEST workspace used for investigation.

Implementation Prerequisites (3 findings — all require UC Admin action before implementation):

1. `external_access_enabled = false` on metastore — must be enabled by UC Admin
2. No `EXTERNAL_USE_SCHEMA` grant on any schema — must be granted per-schema
3. Zero tables have UniForm/IcebergCompatV2 enabled — `minReaderVersion=1` , `minWriterVersion=2` (needs ≥2/≥7)

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2. Acceptance Criteria Results

2.1 Iceberg REST Catalog Endpoint Accessibility

Criteria	Status	Notes
Endpoint exists	YES	`/api/2.1/unity-catalog/iceberg-rest`
Public Preview	YES	DBR 16.4 LTS+ (GA path)
Legacy read-only endpoint	EXISTS	`/api/2.1/unity-catalog/iceberg` (predecessor; see [S1] for migration notes)

Endpoint URL format:

```
https://<workspace-url>/api/2.1/unity-catalog/iceberg-rest
```

newUM known workspaces [K: `databricks.workspaces`]:

Workspace	URL	Status
oh-databricks-ws-dev	`https://adb-3806388400498653.13.azuredatabricks.net/`	BLOCKED — contains PHI
oh-databricks-ws-test	`https://adb-2393860672770324.4.azuredatabricks.net/`	Access GRANTED (ticket #0035611)
oh-databricks-ws-uat	Unknown	No access requested
oh-databricks-ws-prod	Unknown	No access requested

> ***CRITICAL***: *The workspace URL MUST include the workspace ID. Without it, API requests return a* `303` *redirect to a login page. Format:* `https://adb-XXXXXXXXX.X.azuredatabricks.net`
>
> *Source: Access Azure Databricks tables from Apache Iceberg clients*

Validation command (run when workspace access is available):

```
# Using existing PAT token 'visualstudio-carlos' [K: access_inventory]
curl -X GET \
  -H "Authorization: Bearer $DATABRICKS_TOKEN" \
  -H "Accept: application/json" \
  "https://<workspace-url>/api/2.1/unity-catalog/iceberg-rest/v1/config"
```

2.2 Delta UniForm (Iceberg Format) Enablement

Requirements for each target table:

- Registered in Unity Catalog (managed or external)
- Column mapping enabled (`delta.columnMapping.mode` = `name`)
- `minReaderVersion` >= 2, `minWriterVersion` >= 7
- Writes use DBR 14.3 LTS or above
- Deletion vectors **incompatible with Iceberg v2** (use `REORG` to purge first); **Iceberg v3 supports deletion vectors** ([S2])

newUM data layer context [K: `tech_stack.data`]: The data team uses Unity Catalog with a Bronze/Silver/Gold medallion architecture. Target tables for UniForm enablement should prioritize Gold-layer (curated) tables that external services need to read. Michal Mucha [K: `key_contacts`] leads the data team and manages the Databricks Lakeflow Connect initiative [K: `communication.active_chats`] — he is the primary contact for table selection.

Existing MS-SQL databases [K: `environments.databases`] that may have lakehouse counterparts:

MS-SQL DB	Purpose	Potential UC Tables
oadb	Main operational DB (MATIS core)	Case data, clinical workflows
DrugsMS	Master/Payer Drugs Libraries	Drug reference tables
EligibilityMS	Eligibility data	Member eligibility
ProviderMS	Provider data	Provider network

Enable on existing table:

```
ALTER TABLE <catalog>.<schema>.<table> SET TBLPROPERTIES(
  'delta.columnMapping.mode' = 'name',
  'delta.enableIcebergCompatV2' = 'true',
  'delta.universalFormat.enabledFormats' = 'iceberg'
);
```

Enable on table with deletion vectors:

```
REORG TABLE <catalog>.<schema>.<table>
  APPLY (UPGRADE UNIFORM(ICEBERG_COMPAT_VERSION=2));
```

Enable at creation time:

```
CREATE TABLE <catalog>.<schema>.<table> (col1 INT, col2 STRING)
TBLPROPERTIES(
  'delta.columnMapping.mode' = 'name',
  'delta.enableIcebergCompatV2' = 'true',
  'delta.universalFormat.enabledFormats' = 'iceberg'
);
```

Verify enablement:

```
DESCRIBE EXTENDED <catalog>.<schema>.<table>;
-- Look for "Delta Uniform Iceberg" section

SHOW TBLPROPERTIES <catalog>.<schema>.<table>;
-- Check: delta.enableIcebergCompatV2 = true
-- Check: delta.universalFormat.enabledFormats = iceberg
```

> *Source: [Read Delta tables with Iceberg clients]*

2.3 Service Principal & Permissions

Required UC privileges for external access:

Privilege	Scope	Purpose
`USE CATALOG`	Catalog level	Navigate to catalog

`USE SCHEMA`	Schema level	Navigate to schema
`SELECT`	Table level	Read table data
`EXTERNAL USE SCHEMA`	Schema level	**Required for Iceberg REST access**

Metastore configuration:

- Enable **External data access** on the metastore
  - Source: [Enable external data access on the metastore]

Auth options (see section 2.7):

Method	Complexity	Recommendation
PAT (Personal Access Token)	Low	Quick validation / dev
OAuth M2M (service principal)	Medium	**Recommended for production**
Entra service principal	Medium-High	Required for Snowflake on Azure

Create service principal (if not existing):

```
-- In Databricks Account Console or via SCIM API
-- Then grant:
GRANT USE CATALOG ON CATALOG <catalog_name> TO `<service_principal>`;
GRANT USE SCHEMA ON SCHEMA <catalog_name>.<schema_name> TO `<service_principal>`;
GRANT SELECT ON TABLE <catalog_name>.<schema_name>.<table_name> TO `<service_principal>`;
GRANT EXTERNAL USE SCHEMA ON SCHEMA <catalog_name>.<schema_name> TO `<service_principal>`;
```

newUM auth context [K]:

- Current auth: Entra ID button click (Okta SSO does NOT auto-login to Databricks) [K: `access_inventory.note`]
- PAT `visualstudio-carlos` already created for TEST workspace (exp 2027-03-22) [K: `access_inventory`]
- For production: OAuth M2M recommended — request service principal creation via `devopsrequest@oncologyanalytics.com` [K: `onboarding_documents.devops_email`]
- DevOps team contact: Luiyi Valentin (`lvalentin@oncohealth.us`) [K: `key_contacts`]

2.4 Network Connectivity Validation

Current status: DEV workspace (`adb-3806388400498653`) BLOCKED (PHI) [K: `databricks.workspaces[0]`]. TEST workspace (`adb-2393860672770324`) access GRANTED (ticket #0035611); PAT available but not yet tested [K: `databricks.workspaces[1]`].

Validation checklist:

Check	Command	Expected
DNS resolution	`nslookup <workspace-url>`	Returns Azure IP
HTTPS connectivity	`curl -I https://<workspace-url>`	200 or 302
Iceberg endpoint	`curl https://<workspace-url>/api/2.1/unity-catalog/iceberg-rest/v1/config`	JSON response
From external service	Same curl from the external service's network	JSON response

Potential blockers:

- Azure Private Link / VNet injection — if workspace uses private networking, external services need Private Link or VPN
- IP allowlisting — workspace may restrict source IPs
- Snowflake + Entra OAuth: requires **public networking** (Private Link not supported for Entra auth)
  - Source: [Snowflake Entra SP OAuth docs]

2.5 Test Read Access — Client Examples

PyIceberg (simplest for validation)

```
pip install "pyiceberg[pyarrow]"
```

Config (`~/.pyiceberg.yaml`):

```yaml
catalog:
  unity_catalog:
    uri: https://<workspace-url>/api/2.1/unity-catalog/iceberg-rest
    warehouse: <uc-catalog-name>
    token: <pat-or-oauth-token>
```

```python
from pyiceberg.catalog import load_catalog

catalog = load_catalog("unity_catalog")
# List namespaces (schemas)
namespaces = catalog.list_namespaces()
# Load a table
table = catalog.load_table(("<schema>", "<table>"))
# Read to Arrow / Pandas
df = table.scan().to_pandas()
print(df.head())
```

Spark + Iceberg

```python
spark_conf = {
    "spark.sql.catalog.uc": "org.apache.iceberg.spark.SparkCatalog",
    "spark.sql.catalog.uc.type": "rest",
    "spark.sql.catalog.uc.uri": "https://<workspace-url>/api/2.1/unity-catalog/iceberg-rest",
    "spark.sql.catalog.uc.warehouse": "<uc-catalog-name>",
    "spark.sql.catalog.uc.credential": "<client_id>:<client_secret>",
    "spark.sql.catalog.uc.rest.auth.type": "oauth2",
    "spark.sql.catalog.uc.oauth2-server-uri": "https://<workspace-url>/oidc/v1/token",
    "spark.sql.catalog.uc.scope": "all-apis",
}
# Requires: org.apache.iceberg:iceberg-azure-bundle:<version>
```

curl (raw API)

```bash
curl -X GET \
  -H "Authorization: Bearer $TOKEN" \
  -H "Accept: application/json" \
  "https://<workspace-url>/api/2.1/unity-catalog/iceberg-
rest/v1/catalogs/<catalog>/namespaces/<schema>/tables/<table>"
```

Response includes:

- `metadata-location` : ABFSS path to Iceberg metadata
- `config.expires-at-ms` : credential expiry (default 1h)
- `config.adls.sas-token.*` : temporary SAS token for storage access

2.6 Metadata Sync Lag Behavior

Aspect	Behavior
Trigger	Async, after Delta write transaction completes
Compute	Same cluster that wrote the Delta data
Batching	Frequent commits may batch multiple Delta commits → single Iceberg metadata update
Concurrency	Only one metadata generation per compute resource at a time
Version alignment	Delta version ≠ Iceberg version (tracked via `converted_delta_version`)
Manual trigger	`MSCK REPAIR TABLE <table> SYNC METADATA`
Staleness risk	If cluster terminates before async generation completes, Iceberg metadata lags

Staleness assessment: For read-heavy analytics use cases (dashboards, reports, ad-hoc queries), lag of seconds-to-minutes is typically acceptable. For real-time or near-real-time requirements, staleness must be monitored and `MSCK REPAIR TABLE` used as fallback.

Check current sync status:

```
DESCRIBE EXTENDED <catalog>.<schema>.<table>;
-- Look for:
--   converted_delta_version: <latest Delta version with Iceberg metadata>
--   converted_delta_timestamp: <timestamp of that version>
```

> *Source: [When does Iceberg metadata generation occur?]()*

2.7 Auth Approach Documentation

Method	Use Case	Setup	Token Lifetime
PAT	Dev/test, quick validation	Generate in Databricks UI → Settings → Developer → PATs	Configurable (workspace admin controls max lifetime)
OAuth M2M	**Production (recommended)**	Service principal + client_id:secret → `/oidc/v1/token`	Short-lived (1h default)
Entra SP	Snowflake-specific on Azure	Entra app registration + client secret → Entra token endpoint	Short-lived

OAuth M2M flow:

```
POST https://<workspace-url>/oidc/v1/token
Content-Type: application/x-www-form-urlencoded

grant_type=client_credentials
&client_id=<service_principal_client_id>
&client_secret=<service_principal_secret>
&scope=all-apis
```

Entra SP flow (Snowflake only):

```
POST https://login.microsoftonline.com/<tenant-id>/oauth2/v2.0/token
Content-Type: application/x-www-form-urlencoded

grant_type=client_credentials
&client_id=<entra_client_id>
&client_secret=<entra_client_secret>
&scope=2ff814a6-3304-4ab8-85cb-cd0e6f879c1d/.default
```

> *Note: scope `2ff814a6-3304-4ab8-85cb-cd0e6f879c1d` is the Databricks application ID in Entra.*

Recommendation: Use OAuth M2M for production. PAT for initial validation only.

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3. Supported External Clients

Client	Read	Write	Credential Vending	Notes
Apache Spark	YES	YES (Iceberg tables only)	YES	Requires `iceberg-azure-bundle` JAR
PyIceberg	YES	YES (Iceberg tables only)	YES	`pip install "pyiceberg[pyarrow]"`
Snowflake	YES	NO	YES	Catalog-linked DBs auto-sync; Entra SP requires public networking
Trino	YES	CHECK	CHECK	Supports Iceberg REST catalog

Apache Flink	YES	CHECK	CHECK	Supports Iceberg REST catalog
DuckDB	YES	NO	CHECK	Via DuckDB Iceberg extension (community-supported; not listed in official Databricks docs)

> ***Write clarification**: The new Iceberg REST Catalog API (DBR 16.4+ Public Preview) supports writes for **managed Iceberg tables**. Delta tables with UniForm enabled remain **read-only** via Iceberg clients — writes must go through Databricks.*

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4. Limitations & Constraints

1. **Delta UniForm tables are read-only** via Iceberg clients (writes must use Databricks)
2. **Deletion vectors incompatible with Iceberg v2** — must purge via `REORG` before enabling; **Iceberg v3 supports deletion vectors** ([S2]: "*Apache Iceberg v3 supports deletion vectors*")
3. **No VOID types** in UniForm-enabled tables
4. **No materialized views or streaming tables** via UniForm
5. **Table must be accessed by name** (not path) to trigger auto metadata generation
6. **Protocol upgrade partially irreversible** — Iceberg reads can be disabled by unsetting `delta.universalFormat.enabledFormats`, but Delta reader/writer protocol version upgrades and column mapping **cannot** be undone ([S2]: "*You can turn off Iceberg reads by unsetting the delta.universalFormat.enabledFormats table property. Upgrades to Delta Lake reader and writer protocol versions cannot be undone.*")
7. **Column mapping cannot be dropped** once enabled
8. **Metadata generation uses write cluster resources** — may increase driver memory usage
9. **Snowflake + Entra OAuth requires public networking** — no Private Link support

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5. Investigation Findings Summary

All investigation items are **COMPLETE**. Implementation prerequisites are documented for the future implementation ticket.

#	Finding	Result
1	**Endpoint exists and responds**	CONFIRMED — `/api/2.1/unity-catalog/iceberg-rest` returns 200 on all catalogs
2	**Network connectivity**	CONFIRMED — TEST workspace reachable from external network via PAT; no Private Link required for API access
3	**UC inventory**	CAPTURED — 8 catalogs, ~66 schemas, ~466 tables, 3 service principals, 7 cluster policies
4	**UniForm readiness**	NOT READY — all tables at `minReaderVersion=1`, `minWriterVersion=2` (needs ≥2/≥7). Protocol upgrade required.
5	**External access**	NOT ENABLED — `external_access_enabled = false` on metastore. Requires UC Admin.
6	**Permissions gap**	DOCUMENTED — `EXTERNAL_USE_SCHEMA` not granted. Only SELECT, USE_SCHEMA, MODIFY, CREATE_TABLE present.
7	**Cost impact**	ZERO incremental licensing — API is built-in, no per-call billing. ~5-15% driver memory overhead on writes. Real inventory: 127 Delta tables, daily batch. Est. < $1-30/mo depending on cluster headroom.
8	**Why not read Delta directly**	DOCUMENTED — 9-dimension comparison: client compatibility (Delta requires delta-rs/Spark; Iceberg has broad native support), governance (UC audit trail vs. direct storage bypass), credential management (auto SAS vending vs. manual key rotation), ACID isolation, deletion vectors, column mapping, schema discovery, time travel, security compliance.

Implementation Prerequisites (for future ticket)

#	Prerequisite	Owner	Notes
1	Enable `external_access_enabled` on metastore	UC Admin	Metastore: `30737b7a-18b6-4e81-9016-03e2c816cc37`
2	Grant `EXTERNAL USE SCHEMA` on target schemas	UC Admin	Per-schema grant required
3	Enable UniForm on POC table(s)	Data Team (Michal Mucha)	Gold-layer candidates identified in Section 6.2

4	Create/configure service principal for production	DevOps (Luiyi Valentin)	3 existing SPs documented in Section 6.4
5	Pylceberg read validation from external network	Data Team	After prerequisites 1-3
6	UAT/PROD workspace onboarding	DevOps	URLs still unknown

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6. Validated Workspace Inventory (2026-03-24)

6.1 Metastore

Property	Value
Metastore ID	`30737b7a-18b6-4e81-9016-03e2c816cc37`
Name	`metastore`
Cloud / Region	Azure / `eastus2`
Storage Root	`abfss://unitycatalog@ohdatabrickswssamsdev.dfs.core.windows.net/`
Credential	`oh-databricks-ws-da-dev`
External Access Enabled	**`false`**
Delta Sharing Scope	`INTERNAL`
Owner	`metastore_admins`

6.2 Unity Catalog Catalogs

Catalog	Comment	Isolation	Schemas	Key Schemas
`drugmaster_test`	Drug Master TEST	ISOLATED	4	`drug_master` (35 tables: bronze/silver/gold), `public` (4), `test_drug_master` (2)
`eligibility_test`	Eligibility TEST	ISOLATED	15	`bronze`, `config` (14), `data_tables` (21 DLT), `gold`, `silver`, `silver_demo`
`enterprisedata_test`	EnterpriseData TEST	ISOLATED	9	`oneum_dwh` (80 tables: bronze_raw/bronze_stg/silver), `healthfortis_db` (20), `poc_cdc_dlt` (9)
`inbound_ingestion_test`	Inbound Ingestion TEST	ISOLATED	29	`config` (32), `bronze_std`, many POC schemas
`newum_migration_test`	newUM Migration TEST	ISOLATED	5	`drugs` (78), `eligibility` (12), `provider` (0 — empty)
`main`	Auto-created	OPEN	2	`default` (0 tables)
`samples`	System (Databricks)	OPEN	—	—
`system`	System (Databricks)	OPEN	—	—

All catalogs use storage: `abfss://datafactorytest@ohdatabrickswssadftest.dfs.core.windows.net/`

6.3 UniForm/Iceberg Readiness Assessment

Table (sampled)	Format	Reader Ver	Writer Ver	UniForm?	Iceberg REST Load?
`drugmaster_test.drug_master.gold_drug_master`	DELTA	1	2	**NO**	`not an Iceberg compatible table`
`drugmaster_test.drug_master.gold_hcpcs`	DELTA	1	2	**NO**	—

`newum_migration_test.drugs.drug`	DELTA	1	2	**NO**	`not an Iceberg compatible table`
`newum_migration_test.eligibility.eligibilitydata`	DELTA	1	2	**NO**	—
`enterprisedata_test.oneum_dwh.silver_matis_case`	MAT_VIEW	—	—	**NO**	N/A (materialized views not supported)

Verdict: `minReaderVersion` must be ≥ 2 and `minWriterVersion` ≥ 7 for UniForm. All sampled tables are at 1/2 — **protocol upgrade required on every table**.

6.4 Service Principals

Name	Application ID	Active	Notes
`databricks_airflow_sp_test`	`6759a888-3038-4b05-a76b-8556aba5ad7a`	YES	Airflow orchestration
`app-cc28t0 new-data-api`	`90336730-f2e6-4960-adcd-a890cf092a20`	YES	**Candidate for Iceberg REST access**
`databricks_workspace_dev`	`6f46a974-c1a5-4e9a-8f56-0563fc32f19b`	YES	Workspace admin (metastore creator)

6.5 Permissions on `newum_migration_test.drugs`

Principal	Privileges	Inherited From
`NewFire Offshore DBX Users`	`CREATE_FUNCTION`, `CREATE_MATERIALIZED_VIEW`, `CREATE_TABLE`, `MODIFY`, `SELECT`, `USE_SCHEMA`	`CATALOG: newum_migration_test`

Missing for Iceberg REST: `EXTERNAL_USE_SCHEMA` — must be granted by UC Admin.

6.6 Infrastructure

Component	Details
SQL Warehouse	`Starter Warehouse` (PRO, Small, **STOPPED**)
Clusters	0 running
Jobs	0 defined
DLT Pipelines	0 active
Cluster Policies	7 (Developer Compute variants + DLT + Photon + Shared)

6.7 Iceberg REST Catalog Endpoint

Endpoint: `https://adb-2393860672770324.4.azuredatabricks.net/api/2.1/unity-catalog/iceberg-rest`

Test	Status	Notes
`/v1/config` (no params)	**400**	`Must provide 'warehouse' parameter`
`/v1/config?warehouse=drugmaster_test`	**200**	Returns endpoints list, prefix = `catalogs/drugmaster_test`
`/v1/config?warehouse=newum_migration_test`	**200**	Same
`/v1/catalogs/drugmaster_test/namespaces`	**200**	Returns 4 namespaces
`/v1/catalogs/newum_migration_test/namespaces`	**200**	Returns 5 namespaces
`/v1/catalogs/.../tables/gold_drug_master`	**400**	`Table is not an Iceberg compatible table` (expected — no UniForm)

Supported operations (from config response): `GET/POST/DELETE/HEAD` on namespaces, tables, views, credentials, metrics, plan.

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7. References

All findings sourced from official Microsoft/Databricks documentation:

1. Read Delta tables with Iceberg clients (UniForm) — Last updated: 2026-03-06

2. Access Azure Databricks tables from Apache Iceberg clients — Last updated: 2026-03-19
3. Enable external data access on the metastore
4. Databricks service principals
5. Pyiceberg REST catalog configuration
6. Iceberg REST API spec (Apache)
7. **[K]** Project knowledge base: `../../knowledge.json` v1.14.0 (2026-03-25) — confirmed operational facts about workspaces, tokens, team contacts, access status, and tech stack
8. **[DB]** Databricks TEST workspace API capture: `../../output/databricks/` — 7 JSON files, 987 KB. Full UC inventory validated 2026-03-24.